Suspension increases connectedness by one

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:suspension-increases-connectedness' (source lines 801-802, with the enclosing theorem environment on 801-803); the proof environment begins two lines later, so the pairing is by adjacency.

Editorial note. The retained context is the author's own section heading 'Connectedness of suspensions' with its label and the two sentences that announce the aim of the section, so that the counted theorem appears under its original heading.

Editorial note. The retained counter values (chapter 8, section offset 1) let the retained section increment the counter to 2, so the counted theorem prints as Theorem 8.2.1, matching the original numbering.

Editorial note. All five cited labels are recorded as source references rather than pulled into the package, because the retained proof is complete and uses them only as cited criteria and pointers; the counted theorem's own chain of equivalences is retained in full.

Editorial scope. One original result of the HoTT Book, counted once: Suspension increases connectedness by one.

Allowed standard prerequisites (editorial).

- Connectedness and truncatedness. A type A is n-connected when trunc n A is contractible; a type E is n-truncated when it is an n-type; a path type x =_E y of an (n+1)-truncated type is n-truncated.. The statement is phrased with n-connectedness, and the proof uses that the path type of an (n+1)-truncated E is n-truncated at source lines 840-842.
- Suspension as a pushout and truncation of pushouts. The suspension of A is the pushout 1 sqcup^A 1 of the span 1 <- A -> 1, and n-truncation of a pushout cocone is again a pushout among n-truncated types.. The proof starts by rewriting the goal as contractibility of trunc(n+1)(1 sqcup^A 1) (source lines 806-809) and then replaces the pushout by the truncated diagram (source lines 811-821).
- Uniqueness of pushouts and the cocone description of a pushout into an n-type. Two pushouts of the same span among n-types are equivalent; and for a span with both legs out of the same object, a cocone with vertex E into an (n+1)-truncated E is the same as a pair of points of E together with a path between them.. The proof shows that the point 1 is a pushout of the truncated diagram and then applies uniqueness (source lines 823-869).

Source reference. Original section 7.5: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1633–1633. Original result cited by the retained proof but located outside the retained span. It is a navigation pointer: the counted theorem's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original section 6.5: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1710–1710. Pointer to the earlier chapter section where the suspension is defined and identified with a pushout of the span 1 <- A -> 1. The retained sentence cites that section and the proof starts from the identification rather than reproving it.

Source reference. Original section 6.8: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1620–1620. Pointer to the earlier chapter section where suspensions are identified with pushouts; the retained proof starts from that identification rather than reproving it.

Source reference. Original theorem 7.4.12: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1593–1593. Pointer to the separately retained Theorem 7.4.12 (n-truncation commutes with pushouts). The counted theorem in this item is a different proposition, Theorem 8.2.1, whose own author proof is retained in full; this label only records where the cited result lives.

Source reference. Original corollary 7.5.9: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 176–176. Pointer to the original criterion relating connectedness of a type to maps into truncated types; the retained proof uses it as a cited criterion, not as a replacement for its own chain of equivalences.

8.2 Connectedness of suspensions

Recall from original section 7.5 that a type A is called **n-connected** if $\|A\|_n$ is contractible. The aim of this section is to prove that the operation of suspension from original section 6.5 increases connectedness.

Theorem 8.2.1. *If A is n-connected then the suspension of A is $(n+1)$-connected.*

Proof. We remarked in original section 6.8 that the suspension of A is the pushout $\mathbf{1} \sqcup^A \mathbf{1}$, so we need to prove that the following type is contractible:

$$\left\|\mathbf{1} \sqcup^A \mathbf{1}\right\|_{n+1}.$$

By original theorem 7.4.12 we know that $\left\|\mathbf{1} \sqcup^A \mathbf{1}\right\|_{n+1}$ is a pushout in $(n+1)$-Type of the diagram

$$\begin{array}{ccc} \|A\|_{n+1} & \longrightarrow & \|\mathbf{1}\|_{n+1} \\ \downarrow & & \\ \|\mathbf{1}\|_{n+1} & & \end{array}.$$

Given that $\|\mathbf{1}\|_{n+1} = \mathbf{1}$, the type $\left\|\mathbf{1} \sqcup^A \mathbf{1}\right\|_{n+1}$ is also a pushout of the following diagram in $(n+1)$-Type (because both diagrams are equal)

$$\mathscr{D} = \begin{array}{ccc} & & \|A\|_{n+1} \longrightarrow \mathbf{1} \\ & & \Big\downarrow \\ & & \mathbf{1} \end{array}.$$

We will now prove that $\mathbf{1}$ is also a pushout of $\mathscr{D}$ in $(n+1)$-Type. Let E be an $(n+1)$-truncated type; we need to prove that the following map is an equivalence

$$\begin{cases} (\mathbf{1} \to E) & \longrightarrow & \mathsf{cocone}_{\mathscr{D}}(E) \\ y & \longmapsto & (y, y, \lambda u.\, \mathsf{refl}_{y(\star)}) \end{cases}.$$

where we recall that $\mathsf{cocone}_{\mathscr{D}}(E)$ is the type

$$\sum_{(f:\mathbf{1}\to E)} \sum_{(g:\mathbf{1}\to E)} (\|A\|_{n+1} \to (f(\star) =_E g(\star))).$$

The map $\begin{cases} (\mathbf{1} \to E) & \longrightarrow & E \\ f & \longmapsto & f(\star) \end{cases}$ is an equivalence, hence we also have

$$\mathsf{cocone}_{\mathscr{D}}(E) = \sum_{(x:E)} \sum_{(y:E)} (\|A\|_{n+1} \to (x =_E y)).$$

Now A is n-connected hence so is $\|A\|_{n+1}$ because $\left\|\|A\|_{n+1}\right\|_n = \|A\|_n = \mathbf{1}$, and $(x =_E y)$ is n-truncated because E is $(n+1)$-truncated. Hence by original corollary 7.5.9 the following map is an equivalence

$$\begin{cases} (x =_E y) & \longrightarrow & (\|A\|_{n+1} \to (x =_E y)) \\ p & \longmapsto & \lambda z.\, p \end{cases}$$

Hence we have

$$\mathsf{cocone}_{\mathscr{D}}(E) = \sum_{(x:E)} \sum_{(y:E)} (x =_E y).$$

But the following map is an equivalence

$$\begin{cases} E & \longrightarrow & \sum_{(x:E)} \sum_{(y:E)} (x =_E y) \\ x & \longmapsto & (x, x, \mathsf{refl}_x) \end{cases}.$$

Hence

$$\mathsf{cocone}_{\mathscr{D}}(E) = E.$$

Finally we get an equivalence

$$(\mathbf{1} \to E) \simeq \mathsf{cocone}_{\mathscr{D}}(E)$$

We can now unfold the definitions in order to get the explicit expression of this map, and we see easily that this is exactly the map we had at the beginning.

Hence we proved that $\mathbf{1}$ is a pushout of $\mathscr{D}$ in $(n+1)$-Type. Using uniqueness of pushouts we get that $\left\|\mathbf{1} \sqcup^A \mathbf{1}\right\|_{n+1} = \mathbf{1}$ which proves that the suspension of A is $(n+1)$-connected. □